L22_Lecture Scribe

L21_L22 (2).pdf
PDF

L21_L22 (1).pdf
PDF

Generate a lecture scribe suitable for exam preparation. While writing the scribe, explain the definitions and assumptions explicitly, give step-by-step reasoning for definitions, assumptions, proofs and the example step-by-step solutions from the provided context only. All the reasoning should be based solely on the provided context. Make the logical flow explicit so that a student can follow the reasoning during revision. Do not introduce material that is not present in the context.

Here is a **lecture scribe (exam-ready, step-by-step, strictly from your provided slides)**:

📘 Lecture Scribe: Inequalities and Tail Bounds (L21–L22)

---

1. Tail of a Random Variable

Definition

The **tail** of a random variable X is:

$$P(X > x)$$

Step-by-step Understanding

- We focus on **values greater than a threshold** x
- This represents **extreme events (rare cases)**

Why tails matter

- Mean → gives **average behavior (easy)**
- Tail → captures **rare extreme events (hard)**

Examples from context:

- Jobs delayed for long time
- Server overload (too many requests)

Key difficulty

- Computing exact tail probability requires:
  - Summing many small probabilities
- Hence → **complex and computationally expensive**

---

2. Why Tail Bounds?

Problem

Exact computation:

$$P(X \geq k)$$

involves long summations (Binomial/Poisson).

Solution Idea

Instead of exact value:

- Find an **upper bound**
- Easier to compute
- Still guarantees safety

Key Insight

$$P(X \geq k) \leq \text{simple bound}$$

---

3. Markov's Inequality

Definition

For a **non-negative random variable** X:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Assumptions

- $X \geq 0$
- Only **mean** $E[X]$ is known

Proof Idea (Step-by-step)

1. Start with expectation:

$$E[X] = \sum x \cdot P(X = x)$$

2. Focus only on values where $X \geq a$
3. Replace those values by a:

$$E[X] \geq a \cdot P(X \geq a)$$

4. Rearranging:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Interpretation

- Large values are rare
- Uses **only mean** → **very weak bound**

Solved Example (Step-by-step)

Given:

- Average grade = 70
- Threshold = 90

Step 1: Define variable

$$X = \text{student grade}, \quad E[X] = 70$$

Step 2: Apply Markov

$$P(X \geq 90) \leq \frac{70}{90}$$

Step 3: Simplify

$$P(X \geq 90) \leq \frac{7}{9} \approx 0.778$$

Result

- At most **77.8% students** can get A

---

4. Chebyshev's Inequality

Key Idea

Uses:

- Mean μ
- Variance σ^2

Controls:

- **Deviation from mean**

Definition

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Proof Idea

1. Start with deviation:

$$|X-\mu|$$

2. Square it:

$$(X-\mu)^2$$

3. Apply Markov inequality:

$$P((X-\mu)^2 \geq a^2) \leq \frac{E[(X-\mu)^2]}{a^2}$$

4. Use:

$$E[(X-\mu)^2] = \mathrm{Var}(X)$$

Interpretation

- Bounds **how far values go from mean**
- Better than Markov

Solved Example

Given:

- Mean = 70
- Std dev = 5 → Variance = 25
- Threshold = 90

Step 1: Convert condition

$$X \geq 90 \Rightarrow X - 70 \geq 20$$

Step 2: Apply Chebyshev

$$P(X \geq 90) \leq P(|X - 70| \geq 20)$$

Step 3: Substitute

$$\leq \frac{25}{20^2} = \frac{25}{400}$$

Step 4: Simplify

$$= 0.0625$$

Result

- Only **6.25% maximum probability**

---

5. Chernoff Bound

Key Idea

Progression:

- Markov → uses X
- Chebyshev → uses $(X - \mu)^2$
- Chernoff → uses e^{tX}

Insight

- Exponential grows fast
- Amplifies large values
  → tighter tail bound

Core Trick

Transform probability:

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Apply Markov:

$$P(X \geq a) \leq \frac{E[e^{tX}]}{e^{ta}}$$

Final Form

$$P(X \geq a) \leq \min_{t>0} \frac{E[e^{tX}]}{e^{ta}}$$

Key Insight

- Holds for all $t > 0$
- Choose best t → tightest bound

Lower Tail Bound (Derivation Steps)

Goal:

$$P(X \leq a)$$

Step 1: Choose $t < 0$

- Multiplying flips inequality

$$P(X \leq a) = P(tX \geq ta)$$

Step 2: Exponentiate

$$P(e^{tX} \geq e^{ta})$$

Step 3: Apply Markov

$$\leq \frac{E[e^{tX}]}{e^{ta}}$$

Step 4: Minimize over $t < 0$

Final:

$$P(X \leq a) \leq \min_{t<0} e^{-ta} E[e^{tX}]$$

Example (Binomial Case)

Given:

- $X \sim \text{Binomial}(20, 0.5)$
- $\mu = 10$

Goal:

$$P(X = 0)$$

Step 1: Use lower tail

$$(1-\delta)\mu = 0 \Rightarrow \delta = 1$$

Step 2: Apply bound

$$P(X \leq 0) \leq e^{-\mu\delta^2/2}$$

Step 3: Substitute

$$= e^{-10/2} = e^{-5}$$

Result

$$\approx 0.0067$$

→ Very small probability

---

6. Jensen's Inequality

Convex Function Definition

A function $g(x)$ is convex if:

$$g(\alpha x_1 + (1-\alpha)x_2) \leq \alpha g(x_1) + (1-\alpha) g(x_2)$$

Key Interpretation

- Function lies **below straight line**
- Convex combination → weighted average

From Convexity to Expectation

Expectation = weighted average:

$$E[X] = \sum p_i x_i$$

Apply convexity:

$$g(E[X]) \leq E[g(X)]$$

Jensen's Inequality (Final)

$$g(E[X]) \leq E[g(X)]$$

Interpretation

- Apply average first → smaller value
- Apply function first → larger value

Example (Step-by-step)

Given:

- $X = 1$ with 0.5
- $X = 3$ with 0.5
- $g(x) = x^2$

Step 1: Compute mean

$$E[X] = 0.5(1) + 0.5(3) = 2$$

Step 2: Left side

$$g(E[X]) = 2^2 = 4$$

Step 3: Right side

$$E[g(X)] = 0.5(1^2) + 0.5(3^2) = 5$$

Result

$$4 \leq 5$$

---

7. Case Study: System Design Insight

Problem Setup

- X: incoming requests/sec
- C: system capacity
- Failure:

$$X > C$$

Requirement:

$$P(X > C) \leq 0.01$$

Results Using Different Bounds

Method	Info Used	Capacity Required
Markov	Mean only	1,000,000
Chebyshev	Mean + variance	30,000
Chernoff	Full distribution	14,500

Final Insight

Golden Rule:

> More statistical information ⇒ tighter bounds ⇒ lower cost

✅ Final Summary (Exam Ready)

- Tail = probability of extreme events
- Markov:
  - Uses mean
  - Weak bound
- Chebyshev:
  - Uses mean + variance
  - Better bound
- Chernoff:
  - Uses exponential transformation
  - Tightest bound
- Jensen:
  - Convexity ⇒ expectation inequality

If you want, I can convert this into:

- 📄 **1-page revision sheet**
- 🧠 **memory tricks + derivation shortcuts**
- 📝 **important exam questions from this lecture**